ISS 2103 Data Communications & Networks - Exam Map

Built from the course outline (ISE 2103, Mr S. Murove) and 5 past papers: ISS222 May/Jun 2018, ICS/ISE2103 Nov/Dec 2023, ISE2103 Nov/Dec 2024 (SE2024), ISS2103 Nov/Dec 2024 (ISA2024), ISS2103 Nov/Dec 2025. Start here.

Exam format (same every year)

- **3 hours, 100 marks. FIVE questions of 25 marks - answer any FOUR.** Start each question on a fresh page.
- That is **45 minutes per question** - roughly **1.8 minutes per mark**. Leave 10 minutes to check.
- Read all 5 questions first; drop the one with the most parts you can't do. Do your strongest question first.
- Questions mix topics - a single question can have OSI, subnetting and CSMA/CD - so you must cover all topics, not just 4.

Your revision packs

PDF	Topic	Outline weeks	Priority
01	Fundamentals: components, transmission modes, LAN/MAN/WAN, topologies, client-server, protocols	1-2	HIGH
02	OSI & TCP/IP models, IP vs MAC, internetworking	1-2, 5-9	VERY HIGH
03	Physical layer: media, fibre, signals, impairments, modulation, multiplexing, switching, ADSL	1-2	HIGH
04	Data link & MAC: LLC/MAC, CSMA/CD/CA, ALOHA, error detection, flow control, MAC address, bridges, switches, VLANs	3-4	VERY HIGH
05	Network layer: IPv4 classes, SUBNETTING/VLSM, IPv4 & IPv6 headers, IPv6, SLAAC vs DHCPv6, DHCP	5-7	VERY HIGH
06	Routing (RIP, EIGRP, OSPF), congestion control, QoS, SDN, ISPs, SLA, WAN protocols	5-7	HIGH
07	Transport & application: TCP vs UDP, handshake, DNS, troubleshooting, RPC, compression, FTP, email	8-11	MEDIUM-HIGH
08	Security & emerging tech: threats, malware, DES vs AES, KDC, CA, VPN	8-9, 12	MEDIUM

Where every past question falls (topic x paper)

Topic	2018	2023	SE 2024	ISA 2024	2025
01 Fundamentals	Q1a, Q1c	Q4e	Q1a, Q2e-f, Q3d-e, Q4a-b, Q4d-e, Q5c-d	Q2c, Q3b	Q1b, Q2a, Q2c
02 OSI / TCP-IP	Q5b	Q1c	Q1e, Q3b	Q3c	Q1a, Q5a, Q5c

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Topic	2018	2023	SE 2024	ISA 2024	2025
03 Physical / switching	-	Q1e, Q3a-b, Q5b	Q1b, Q1d, Q2a, Q2c, Q3c, Q4c, Q5b	Q2d, Q5b	Q2b, Q3a
04 Data link / MAC	Q1b, Q3b, Q4b, Q4d, Q5c-d	Q4a-b, Q5a	Q1b.ii, Q2b, Q3a, Q4f	Q1b-c, Q4a, Q4c	Q4c, Q5b, Q5d
05 IP / subnetting / IPv6	Q1d, Q3a, Q4a	Q1a-b, Q1d, Q3c, Q5e	Q1b.iv	Q1a, Q1d, Q2b, Q3a, Q5d	Q3b, Q4a-b
06 Routing / QoS / WAN	Q4c	Q2a-e, Q3d-e, Q5d	Q1c	Q4b, Q4d, Q5c	Q3c
07 Transport / application	Q2a-c, Q3c, Q5a	Q5c	Q5e	Q5a	-
08 Security	-	Q4c-d	Q2d, Q5a	Q2a	-

EXAM TIP: Your paper is the **ISS (Information Security & Assurance)** version - the ISA2024 and 2025 papers are the closest to what you will get. They are heavy on OSI, subnetting, switching, CSMA/CD, LLC/MAC, IPv6/SLAAC, routing and congestion.

Most repeated questions - the 'bankers'

#	Question	Times seen	Pack
1	Subnetting / VLSM calculation	5 of 5 papers (7 parts)	05
2	OSI layers + functions (often with diagram); OSI vs TCP/IP	5 of 5	02
3	Network topologies (describe 3-4; ring vs bus)	3 papers	01
4	Circuit vs packet (vs message) switching + examples	4 papers	03
5	CSMA/CD (and CSMA/CD vs CSMA/CA)	4 papers	04
6	TCP vs UDP / how UDP sends data / why TCP reliable	5 papers	07
7	IPv4 header fields (TTL, protocol, version, DS, addresses)	2 papers	05
8	IPv6: shorten addresses; advantages; header	3 papers	05
9	SLAAC vs DHCPv6	2 papers (the last two!)	05
10	LLC and MAC sub-layers	2 papers	04
11	Transmission media types; fibre optic	4 papers	03
12	Error detection (parity, CRC, checksum; single-bit vs burst)	2 papers	04
13	Client-server vs P2P	2 papers	01
14	Routing: DV vs LS, RIP/RIPv2/EIGRP/OSPF	2 papers	06
15	Multiplexing (concept, FDM vs TDM)	2 papers	03
16	LAN/MAN/WAN; categories of networks	2 papers	01

Likely 'new' questions (in the outline but not yet asked much)

- Sliding window protocols (Go-Back-N vs Selective Repeat) - Pack 04.
- FDDI, IEEE 802 standards overview, channel allocation - Pack 04.
- Remote Procedure Call; data compression techniques; session layer design issues - Pack 07.
- FTP, email protocols (SMTP/POP3/IMAP), virtual terminals - Pack 07.
- ISDN; terminal handling; backbone & local-access network design - Packs 03 & 01.
- Emerging technologies (5G, IoT, cloud, SDN) - Pack 08.
- Lab content: network commands (ipconfig, ping, tracert, netstat), straight-through vs crossover cables, Packet Tracer - Pack 07.

How to write answers that score

Command word	What the examiner wants
List / state / name	Short points only, one per mark
Define / explain the term	Clear definition + one example (or how it works)
Describe / explain	Definition + how it works step by step + example; a diagram earns marks
Differentiate / distinguish / compare	A **table** with matching rows (e.g. 4 rows for 4 marks); 'compare' also wants similarities
With the aid of a diagram	The diagram must be **labelled** - usually ~half the marks
Calculate / show working	Show every step: binary mask, borrowed bits, block size, then a results table
Discuss	Several points, each explained, with advantages/disadvantages or examples

One-page last-minute facts

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OSI 7->1: Application, Presentation, Session, Transport, Network, Data Link, Physical
          (All People Seem To Need Data Processing)
PDUs: data / segment / packet / frame / bits      Devices: L1 hub+repeater, L2 switch+bridge, L3 router
Delivery: L2 hop-to-hop (MAC), L3 host-to-host (IP), L4 process-to-process (port)
TCP/IP: Application | Transport | Internet | Network Access

Hosts = 2^h - 2   Subnets = 2^borrowed   Block = 256 - mask octet
/24 254  /25 126  /26 62  /27 30  /28 14  /29 6  /30 2
Class A 1-126 /8, B 128-191 /16, C 192-223 /24.  Private: 10/8, 172.16-31, 192.168/16
IPv4 header 20 B (TTL, Protocol 6=TCP 17=UDP 1=ICMP).  IPv6: 128-bit, header 40 B, hop limit, flow label
IPv6 shorten: drop leading zeros, '::' once for the longest zero run
SLAAC: host builds address from RA prefix (stateless).  DHCPv6: server assigns & tracks (stateful)
DHCP: Discover, Offer, Request, Ack (UDP 67/68)

CSMA/CD wired (detect, jam, backoff) vs CSMA/CA wireless (avoid, RTS/CTS, ACK)
Pure ALOHA 18.4% (vulnerable 2T), Slotted ALOHA 36.8% (vulnerable T)
Errors: single-bit -> parity; burst -> CRC; checksum in IP/TCP/UDP.  FEC = receiver corrects (Hamming)
Hamming distance = number of differing bits; detect s: d=s+1; correct t: d=2t+1
MAC = 48-bit: 24-bit OUI + 24-bit NIC.  ARP: IP -> MAC (broadcast request, unicast reply)

Circuit = dedicated path (phone, ISDN). Packet = shared, independent packets (IP, X.25, Frame Relay)
FDM = frequency bands (analogue), TDM = time slots (digital), WDM = light wavelengths
Media: twisted pair, coax, fibre (guided) | radio, microwave, infrared (unguided)
Impairments: attenuation, distortion, noise

RIP: DV, hop count (max 15), 30 s updates, v2 = classless + multicast 224.0.0.9 + auth
EIGRP: advanced DV, DUAL, bandwidth+delay, successor/feasible successor.  OSPF: LS, Dijkstra, cost
QoS: bandwidth, latency, jitter, packet loss.  VoIP: <150 ms, <30 ms jitter, <1% loss
TCP: SYN -> SYN/ACK -> ACK.  UDP: 8-byte header, no connection, no ACK.  DNS UDP 53
DES 56-bit key 64-bit block 16 rounds; AES 128/192/256-bit key 128-bit block 10/12/14 rounds
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